

EC 111: Principles of Microeconomics

Lecture Notes, Chapter 4: The Market Forces of Supply and Demand

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4 The Market Forces of Supply and Demand

Supply and demand are the two words economists use most, and this chapter is where the tools behind them are built. The model answers four questions: what determines how much buyers want to buy, what determines how much sellers want to sell, how a price emerges from their interaction, and how that price responds when circumstances change. Almost every later chapter uses this machinery, so the time invested here pays off repeatedly.

4.1 Markets and competition

Definition 4.1 (Market). A **market** is a group of buyers and sellers of a particular good or service. The buyers as a group determine the demand for the product; the sellers as a group determine its supply.

Markets take many forms. Some are organized (a stock exchange, a farmers' market); most are not (the market for haircuts in Waltham consists of a few dozen shops and everyone who walks into them). What matters for the model is not the physical setting but how much influence any single participant has over the price.

Definition 4.2 (Competitive market). A **competitive market** has many buyers and many sellers, each so small relative to the market that no one of them can affect the market price. A **perfectly competitive market** adds two assumptions:

- (i) the goods offered by different sellers are identical, and
- (ii) buyers and sellers are so numerous that each one is a **price taker**: they treat the market price as given and decide only how much to buy or sell at that price.

Few real markets meet both assumptions exactly (wheat and other commodities come closest), but the model of perfect competition is the right starting point. It is the simplest case, it captures the essential logic that reappears in more complicated markets, and later chapters relax the assumptions one at a time.

4.2 Demand

4.2.1 Quantity demanded and the law of demand

Definition 4.3 (Quantity demanded). The **quantity demanded** of a good is the amount buyers are willing *and able* to purchase at a given price. Both parts matter: wanting a car is not demand; wanting a car and having the money to buy one at the going price is.

Definition 4.4 (Law of demand). Other things being equal, the quantity demanded of a good falls when its price rises and rises when its price falls.

The phrase *other things being equal* (*ceteris paribus*) is doing real work. The law of demand describes how quantity responds to price when everything else that might influence buyers, such as their incomes, the prices of other goods, and their tastes, is held fixed. Those other things are addressed below under shifts of the demand curve.

4.2.2 The demand schedule and the demand curve

Definition 4.5 (Demand schedule and demand curve). A **demand schedule** is a table listing the quantity demanded at each price, other things equal. A **demand curve** is the graph of that relationship, with price on the vertical axis and quantity on the horizontal axis.

Example 4.1 (A market demand schedule). Table 1 gives the demand schedule for a good sold at prices from \$0 to \$50. The quantity demanded falls by 10 units for every \$10 increase in price, so the schedule can also be written as the equation $Q_d = 50 - P$. Figure 1 plots the six points and connects them.

Price (\$)	Quantity demanded
0	50
10	40
20	30
30	20
40	10
50	0

Table 1: Demand schedule: $Q_d = 50 - P$.

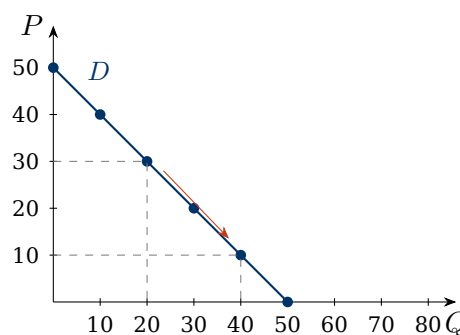


Figure 1: The demand curve. A price cut from \$30 to \$10 raises quantity demanded from 20 to 40: a movement along the curve.

The demand curve slopes downward: that is the law of demand in graphical form. A change in the price of the good moves buyers from one point on the curve to another. It does not move the curve.

4.2.3 Individual demand and market demand

Definition 4.6 (Market demand). The **market demand** for a good is the sum of the individual demands of all buyers in the market. At each price, the market quantity demanded is the sum of the quantities demanded by every buyer. Graphically, the market demand curve is the *horizontal* sum of the individual demand curves.

Example 4.2 (Two buyers). Suppose the market in Table 1 has only two buyers, A and B, with the individual schedules in Table 2. At a price of \$20, A wants 18 units and B wants 12, so the market quantity demanded is 30. Repeating this at every price reproduces the market schedule. In equation form, $Q_A = 30 - 0.6P$ and $Q_B = 20 - 0.4P$, so $Q_d = Q_A + Q_B = 50 - P$.

Price (\$)	Buyer A	Buyer B	Market Q_d
0	30	20	50
10	24	16	40
20	18	12	30
30	12	8	20
40	6	4	10
50	0	0	0

Table 2: Market demand is the horizontal sum of individual demands.

A two-buyer market violates the “many buyers” assumption, but the point holds with two buyers or two million: add quantities across buyers at each price.

4.2.4 Shifts of the demand curve

The demand curve is drawn holding fixed everything other than the good’s own price. When one of those other things changes, the quantity demanded changes *at every price*, and the whole curve moves.

Definition 4.7 (Increase and decrease in demand). An **increase in demand** is a rightward shift of the demand curve: buyers want more at every price. A **decrease in demand** is a leftward shift: buyers want less at every price.

Five variables shift the demand curve. Table 3 lists them with the direction of the shift.

Definition 4.8 (Normal and inferior goods). A good is **normal** if, other things equal, an increase in income raises the demand for it. A good is **inferior** if an increase in income lowers the demand for it. Most goods are normal; instant noodles, bus rides, and secondhand furniture are common examples of inferior goods.

Definition 4.9 (Substitutes and complements). Two goods are **substitutes** if an increase in the price of one raises the demand for the other (pizza and burgers, Coke and Pepsi, streaming and the movie theater). Two goods are **complements** if an increase in the price of one lowers the demand for the other (smartphones and apps, printers and ink, tuition and textbooks).

Shifter	Change	Effect on demand
Number of buyers	More buyers enter the market	Increases (shifts right)
Income	Income rises, good is normal	Increases
	Income rises, good is inferior	Decreases (shifts left)
Price of a related good	Price of a substitute rises	Increases
	Price of a complement rises	Decreases
Tastes	Preferences move toward the good	Increases
Expectations	Buyers expect a higher future price or higher future income	Increases (today)

Table 3: Demand shifters. Each row reversed gives the opposite shift.

Whether a good is normal or inferior, and whether two goods are substitutes or complements, is an empirical matter about buyers' behavior, not a property of the goods themselves. Frozen pizza may be inferior for one group of buyers and normal for another.

Key point

A change in the good's own price causes a **change in quantity demanded**: a movement along a fixed demand curve. A change in any other determinant causes a **change in demand**: a shift of the entire curve. The rule of thumb: if the variable that changed is on one of the axes, move along the curve; if it is not on either axis, shift the curve.

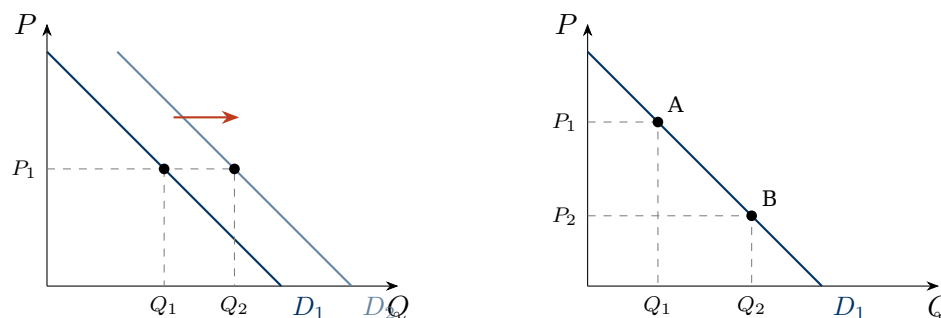


Figure 2: Left: an increase in demand shifts the curve right; at the unchanged price P_1 , buyers want Q_2 instead of Q_1 . Right: a fall in the good's own price moves buyers from A to B along the same curve.

In class

Spotify sells the same subscription at the same price in every country it serves. Draw the demand curve for Spotify subscriptions and show what each event does to it.

- (a) Spotify launches in twenty more countries without changing its price.
- (b) A global recession lowers incomes everywhere.
- (c) Apple Music cuts its price.
- (d) Smartphones become cheaper.
- (e) Spotify starts a campaign promoting local artists in developing countries.
- (f) Prominent economists warn that a recession is coming.
- (g) Spotify eliminates its free tier, so new users must pay the subscription price.

Answers. (a) More buyers: demand shifts right. (b) Income falls; if a subscription is a normal good, demand shifts left. The question of whether Spotify is normal or inferior has no answer from theory alone; it depends on how subscribers actually respond, and a plausible case can be made either way (people may cut subscriptions, or may substitute cheap streaming for expensive concerts). (c) Apple Music is a substitute; a lower substitute price shifts Spotify demand left. (d) Phones are a complement; cheaper phones shift demand right. (e) Tastes: if the campaign makes the service more attractive, demand shifts right. (f) Expected future income falls; demand shifts left today. (g) For the users affected, the price they face rises from zero to the subscription price. That is a change in the good's own price: a movement along the demand curve, not a shift.

4.3 Supply

4.3.1 Quantity supplied and the law of supply

Definition 4.10 (Quantity supplied). The **quantity supplied** of a good is the amount sellers are willing and able to sell at a given price.

Definition 4.11 (Law of supply). Other things being equal, the quantity supplied of a good rises when its price rises and falls when its price falls.

A higher price makes each unit sold more profitable, so existing sellers produce more and marginal producers find it worth entering. The supply curve therefore slopes upward.

4.3.2 The supply schedule and the supply curve

Definition 4.12 (Supply schedule and supply curve). A **supply schedule** is a table listing the quantity supplied at each price, other things equal. A **supply curve** is the graph of that relationship.

Example 4.3 (A market supply schedule). Table 4 gives the supply schedule for the same good as Table 1. Quantity supplied rises by 15 units for every \$10 increase in price: $Q_s = 1.5P$. Figure 3 plots it.

Price (\$)	Quantity supplied
0	0
10	15
20	30
30	45
40	60
50	75

Table 4: Supply schedule: $Q_s = 1.5P$.

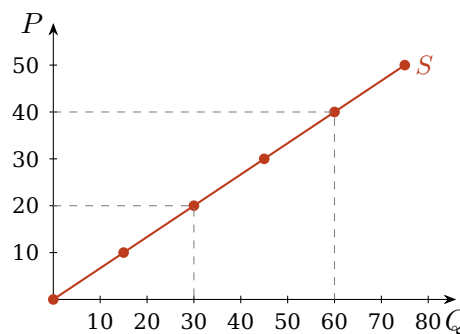


Figure 3: The supply curve. A price rise from \$20 to \$40 raises quantity supplied from 30 to 60: a movement along the curve.

4.3.3 Individual supply and market supply

Definition 4.13 (Market supply). The **market supply** of a good is the sum of the individual supplies of all sellers. At each price, add the quantities supplied by every seller; graphically, sum the individual supply curves horizontally.

Example 4.4 (Two sellers). If the market in Table 4 has two sellers with $Q_1 = P$ and $Q_2 = 0.5P$, then at \$20 seller 1 offers 20 units and seller 2 offers 10, for a market quantity supplied of 30. Summing at every price gives $Q_s = 1.5P$.

4.3.4 Shifts of the supply curve

Definition 4.14 (Increase and decrease in supply). An **increase in supply** is a rightward shift of the supply curve: sellers offer more at every price. A **decrease in supply** is a leftward shift: sellers offer less at every price.

Shifter	Change	Effect on supply
Input prices	Wages, raw materials, energy become cheaper	Increases (shifts right)
Technology	A cost-saving improvement in production	Increases
Number of sellers	More firms enter the market	Increases
Expectations	Sellers expect a higher future price (storable goods)	Decreases today (shifts left)

Table 5: Supply shifters. Each row reversed gives the opposite shift.

Input prices and technology work through cost: anything that lowers the cost of producing a unit makes production more profitable at every price, so sellers offer more. The expectations row runs the opposite way from the demand table: a seller who expects a higher price next month holds inventory back today, reducing current supply.

Common mistake

Two errors recur on exams.

- Describing a supply shift as “up” or “down.” A supply curve that moves up is a *decrease* in supply (less offered at each price). Always describe shifts as right (increase) or left (decrease), reading horizontally at a fixed price.
- Putting a demand shifter on the supply curve. A fall in the price of orange juice affects the *demand* for apple juice (they are substitutes for buyers). It does nothing to the supply of apple juice, because it does not change the cost of making apple juice.

Key point

A change in the good’s own price causes a **change in quantity supplied**: a movement along the supply curve. A change in input prices, technology, the number of sellers, or expectations causes a **change in supply**: a shift of the curve.

4.4 Supply and demand together**4.4.1 Equilibrium**

Definition 4.15 (Equilibrium). A market is in **equilibrium** when the price has reached the level at which quantity supplied equals quantity demanded. That price is the **equilibrium price** (or market-clearing price) and the common quantity is the **equilibrium quantity**. On the graph, equilibrium is the intersection of the supply and demand curves.

Example 4.5 (Equilibrium from the schedules). Table 6 puts the demand and supply schedules side by side. At \$20, quantity demanded and quantity supplied are both 30. That is the only price at which they match, so the equilibrium is $P^* = 20$, $Q^* = 30$. Figure 4 shows the same thing graphically.

Price (\$)	Q_d	Q_s	$Q_s - Q_d$
0	50	0	-50
10	40	15	-25
20	30	30	0
30	20	45	25
40	10	60	50
50	0	75	75

Table 6: Negative entries in the last column are shortages; positive entries are surpluses.

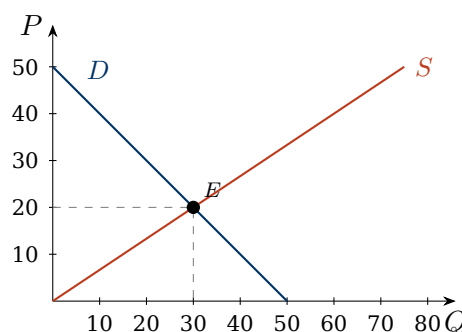


Figure 4: Equilibrium at $P^* = 20$, $Q^* = 30$.

4.4.2 Surplus and shortage

What if the price is not at its equilibrium level? The last column of Table 6 answers this.

Definition 4.16 (Surplus). A **surplus** (excess supply) exists when the quantity supplied exceeds the quantity demanded. It arises when the price is above the equilibrium price.

Definition 4.17 (Shortage). A **shortage** (excess demand) exists when the quantity demanded exceeds the quantity supplied. It arises when the price is below the equilibrium price.

Example 4.6 (Surplus at \$40, shortage at \$10). At $P = 40$, sellers offer 60 units but buyers want only 10: a surplus of 50. Sellers holding unsold inventory cut the price. As the price falls, quantity demanded rises (down along D) and quantity supplied falls (down along S), so the surplus shrinks until it disappears at \$20.

At $P = 10$, buyers want 40 units but sellers offer only 15: a shortage of 25. Too many buyers chase too few units, and sellers can raise the price without losing customers. As the price rises, quantity demanded falls and quantity supplied rises, until the shortage is gone at \$20. Figure 5 shows both cases.

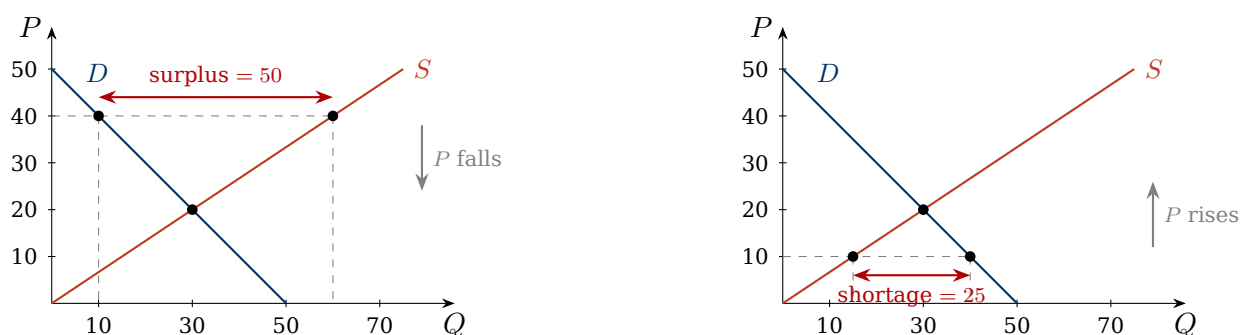


Figure 5: Left: at \$40 a surplus of 50 pushes the price down. Right: at \$10 a shortage of 25 pushes it up. Both end at \$20.

Key point

Law of supply and demand. The price of any good adjusts to bring the quantity supplied and the quantity demanded into balance. Once the market reaches equilibrium, there is no further pressure on the price in either direction. How fast this happens varies across markets, but in most well-functioning markets surpluses and shortages are temporary.

4.4.3 Equilibrium from equations

When demand and supply are given as equations, equilibrium is the price that solves $Q_d = Q_s$.

Example 4.7 (Solving for equilibrium). Let $Q_d = 1200 - 50P$ and $Q_s = 200 + 50P$. Setting them equal,

$$1200 - 50P = 200 + 50P \implies 1000 = 100P \implies P^* = 10.$$

Substituting back, $Q^* = 1200 - 50(10) = 700$; checking with supply, $200 + 50(10) = 700$. Always verify with the second equation, since an arithmetic slip in the first will show up as a mismatch.

The schedules of Table 6 work the same way: $50 - P = 1.5P$ gives $P^* = 20$ and $Q^* = 30$.

To find a surplus or shortage at a given price, evaluate both equations at that price and subtract. At $P = 14$ in the example, $Q_d = 500$ and $Q_s = 900$: a surplus of 400. At $P = 6$, $Q_d = 900$ and $Q_s = 500$: a shortage of 400.

4.5 Three steps to analyzing a change in equilibrium

An event that shifts demand or supply changes the equilibrium. To predict how, follow three steps.

Key point

1. Decide whether the event shifts the demand curve, the supply curve, or both.
2. Decide whether each affected curve shifts right or left.
3. Draw the diagram and compare the new equilibrium with the old one: what happened to P and Q ?

Step 1 requires knowing the shifter lists (Tables 3 and 5). Ask: does the event change what buyers want at a given price, or what it costs sellers to produce? Step 2 requires the direction. Step 3 is mechanical once the first two are right.

Example 4.8 (The electric car market: three cases). Assume the market for electric cars starts in equilibrium at (Q_1, P_1) .

Case A: the price of oil rises. Gasoline cars run on oil; electric cars do not. For a buyer, a gasoline car and an electric car are substitutes, and the higher oil price makes gasoline cars more expensive to own. Step 1: demand shifts (this is a related-good price on the buyer side; the cost of building an electric car is unchanged). Step 2: demand shifts right. Step 3: at the old price there is a shortage, the price is bid up, and the new equilibrium has a higher price and a higher quantity (Figure 6, left).

Case B: engine technology improves. A cheaper way to build the motor lowers production cost. Step 1: supply shifts. Step 2: supply shifts right. Step 3: at the old price there is a surplus, the price falls, and the new equilibrium has a lower price and a higher quantity (Figure 6, center).

Case C: both at once. Both curves shift right. Both shifts raise quantity, so Q rises unambiguously. The demand shift pushes the price up and the supply shift pushes it down, so the net effect on P depends on which shift is larger. Without that information, the effect on price is **ambiguous** (Figure 6, right, drawn with the supply shift larger, so that P falls).

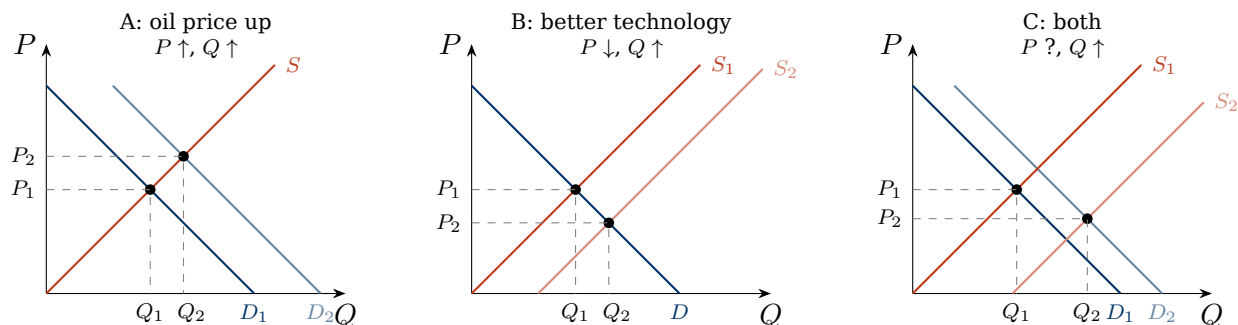


Figure 6: The electric car market. In case C the direction of P depends on the relative sizes of the shifts.

The same logic covers every combination. Table 7 summarizes. When only one curve shifts, both P and Q are determined. When both shift, one of the two is determined and the other is ambiguous.

	No change in supply	Supply increases	Supply decreases
No change in demand	P same, Q same	$P \downarrow, Q \uparrow$	$P \uparrow, Q \downarrow$
Demand increases	$P \uparrow, Q \uparrow$	P ambiguous, $Q \uparrow$	$P \uparrow, Q$ ambiguous
Demand decreases	$P \downarrow, Q \downarrow$	$P \downarrow, Q$ ambiguous	P ambiguous, $Q \downarrow$

Table 7: Effects of shifts on equilibrium price and quantity.

Note

When the problem says both curves shift but not by how much, the correct answer for the ambiguous variable is “cannot be determined,” not a guess. If the problem does specify which shift is larger, the ambiguity disappears: in case C, a larger demand shift means P rises, a larger supply shift means P falls.

Example 4.9 (Eggs, 2022–2025). The average U.S. price of a dozen Grade A large eggs was \$1.93 in January 2022, \$4.82 in January 2023, and \$6.23 in March 2025 (BLS average price data). Apply the three steps. *Which curve?* Highly pathogenic avian influenza killed or forced the culling of more than 40 million laying hens in 2022, with further waves in late 2024 and early 2025 (USDA APHIS). Fewer hens is a change in the number of sellers and in the cost of producing eggs: it affects sellers, not buyers, so it is the *supply* curve. *Which way?* Left. *Compare equilibria.* At the old price the quantity demanded exceeds the quantity supplied, and the shortage bids the price up until the market clears again: the price rises and the quantity falls.

Two things are worth noticing. First, nobody chose a price of \$6; the shortage produced it. Second, the same higher price that rations the remaining eggs among the buyers who value them most also tells producers to rebuild their flocks, which is what eventually brings the price back down.

In class

For each statement, draw the shift in the named market that is consistent with it and identify the shifter.

- (a) Healthcare: “When there is an influx of immigrants, the price of healthcare rises.”
- (b) Electric cars: “When the price of electric cars is expected to rise in the near future, the current price of electric cars rises.”
- (c) Laptops, then used desktops: “When a technological advance improves the features of laptops, the price of laptops rises and the price of used desktops falls.”

Answers. (a) More buyers: demand for healthcare shifts right; P and Q rise. (b) Two forces point the same way. Buyers who expect a higher price buy now (demand right). Sellers who expect a higher price hold cars back (supply left). Either alone raises today's price; together they raise it for sure and leave Q ambiguous. Drawing only the demand shift is enough to match the statement. (c) This is a two-market problem. In the laptop market, the advance improves the product rather than lowering its cost, so it works through tastes: demand for laptops shifts right and the laptop price rises. (If instead the advance lowered production cost, supply would shift right and the price would fall, which contradicts the statement; the statement tells you which interpretation is meant.) In the used-desktop market, laptops are a substitute, and the better laptop draws buyers away: demand for used desktops shifts left, and their price falls.

4.6 How prices allocate resources

Chapter 1 claimed that markets are usually a good way to organize economic activity. This chapter shows the mechanism. In a market economy nobody decides how many electric cars to build or who gets them. The price does. When demand rises, the higher price signals producers to build more and rations the cars among the buyers who value them most; when a new technology cuts costs, the lower price spreads the gain to buyers and pulls resources into the industry. Prices are the signals that guide every decision in the economy, and the equilibrium price is the one that makes those decisions consistent: what buyers choose to consume equals what sellers choose to produce.

Friedrich Hayek put the point sharply in 1945. Suppose a source of supply of a raw material disappears; it does not matter, for the argument, whether it was destroyed or whether a new use for it was found.

All that the users of tin need to know is that some of the tin they used to consume is now more profitably employed elsewhere and that, in consequence, they must economize tin. ... The marvel is that in a case like that of a scarcity of one raw material, without an order being issued, without more than perhaps a handful of people knowing the cause, tens of thousands of people whose identity could not be ascertained by months of investigation, are made to use the material or its products more sparingly; i.e., they move in the right direction.

— F. A. Hayek, “The Use of Knowledge in Society,” *American Economic Review*, 1945

The price carries the information. No one in the chain needs to know why tin became scarce, and no central office needs to collect what each of them knows. That is the sense in which a market “organizes” activity, and it is also the reason that later chapters, which study cases where prices fail to carry the right information, matter so much.

That is the idea behind the invisible hand. The rest of the course studies when this mechanism works well and when it does not.

4.7 Exercises

Exercise 4.1. The market for pizza slices on campus has the following schedules.

Price (\$)	Q_d	Q_s
1	120	0
2	100	40
3	80	80
4	60	120
5	40	160
6	20	200

- Find the equilibrium price and quantity.
- At a price of \$5, is there a surplus or a shortage, and how large? What happens to the price?
- At a price of \$2, is there a surplus or a shortage, and how large?
- A new dorm opens and quantity demanded rises by 60 slices at every price. Write the new demand schedule and find the new equilibrium.

Exercise 4.2. Demand and supply for a good are $Q_d = 1200 - 50P$ and $Q_s = 200 + 50P$.

- Find the equilibrium price and quantity.
- Find the surplus or shortage at $P = 16$ and at $P = 4$.
- Income rises and the good is normal; demand becomes $Q_d = 1500 - 50P$. Find the new equilibrium. Then explain, using the words *shortage* and *movement along the supply curve*, how the market gets there.

Exercise 4.3. Apply the three-step method to each event and state the effect on equilibrium price and quantity.

- A hurricane destroys a large share of the Hudson Valley apple crop. Analyze the market for apples. Then analyze the market for apple cider.
- A scientific report finds that fish contain dangerous levels of mercury. At the same time, the price of diesel fuel used by fishing boats falls. Analyze the market for fish.

- Exercise 4.4.** (a) Honey and jam are substitutes. The price of honey rises. What happens in the market for honey? What happens in the market for jam, and to the price of jam?
- (b) Retirees are the main buyers of housing in Florida. Which of the following shifts the demand curve for Florida housing to the right: (i) the aging of the baby boom generation, which raises the number of people reaching retirement; (ii) higher homeowners' insurance premiums; (iii) high current real-estate prices; (iv) stricter building codes that raise construction costs. Explain what each of the others does instead.

Answers to exercises

Exercise 4.1. (a) $Q_d = Q_s = 80$ at $P = 3$, so $P^* = 3$, $Q^* = 80$. (b) At \$5, $Q_s = 160$ and $Q_d = 40$: a surplus of 120. Sellers with unsold slices cut the price; it falls toward \$3. (c) At \$2, $Q_d = 100$ and $Q_s = 40$: a shortage of 60. The price rises. (d) New Q_d at prices \$1 to \$6: 180, 160, 140, 120, 100, 80. It equals Q_s at \$4 ($120 = 120$), so the new equilibrium is $P^* = 4$, $Q^* = 120$. More buyers, demand right, P and Q both up.

Exercise 4.2. (a) $1200 - 50P = 200 + 50P$ gives $P^* = 10$, $Q^* = 700$. (b) At $P = 16$: $Q_d = 400$, $Q_s = 1000$, surplus of 600. At $P = 4$: $Q_d = 1000$, $Q_s = 400$, shortage of 600. (c) $1500 - 50P = 200 + 50P$ gives $P^* = 13$, $Q^* = 850$ (check: $200 + 650 = 850$). At the old price of \$10, quantity demanded is now $1500 - 500 = 1000$ while quantity supplied is still 700: a shortage of 300. The shortage pushes the price up. As it rises, sellers move up along the unchanged supply curve from 700 toward 850, and buyers move up along the new demand curve from 1000 toward 850. The shortage closes at \$13.

Exercise 4.3. (a) Apples: the hurricane reduces the number of sellers able to supply, and raises the cost of supplying, apples. Supply shifts left; P rises, Q falls. Cider: apples are an input, and their price has risen. Supply of cider shifts left; the price of cider rises and the quantity falls. (b) Fish: the report changes tastes; demand shifts left. Cheaper diesel lowers an input price; supply shifts right. Both shifts lower the price, so P falls. The demand shift lowers Q and the supply shift raises it, so Q is ambiguous.

Exercise 4.4. (a) Honey: a change in the good's own price is a movement along the honey demand curve, not a shift; quantity of honey demanded falls. Jam: honey is a substitute and has become more expensive, so demand for jam shifts right; the price and quantity of jam both rise. (b) Only (i): more buyers shift demand right. (ii) Insurance is a complement to owning a home; higher premiums shift demand left. (iii) The current price of housing is on the vertical axis; it determines a point on the demand curve, not its position. (iv) Construction cost is a seller's cost; it shifts supply left, not demand.